

# Gowtham D

## AI/ML Engineer | 9282355652 | gowthamdodda.92@gmail.com

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### SUMMARY:

#### Professional Summary

- AI Engineer with over 6 years of experience building and deploying scalable machine learning and generative AI solutions.
- Utilized AI-assisted development tools to design key platform features and deliver production-ready, end-to-end AI systems.
- Interpreted and worked with complex production-grade Python code to integrate AI-driven insights into IT operations seamlessly.
- Developed lightweight Python scripts to support data ingestion, process automation, and basic workflow orchestration.
- Built and deployed machine learning models aimed at improving IT incident management efficiency.
- Analyzed large volumes of operational data to uncover predictive insights and enhance system performance using AI methodologies.
- Maintained high scalability and reliability of AI models operating in large-scale production environments.
- Provided mentorship to junior engineers, offering guidance on AIOps practices and AI engineering standards.
- Assessed and selected appropriate AI tools and frameworks for modern AIOps implementations.
- Created detailed documentation for AI models, workflows, and methodologies to support knowledge sharing and compliance.
- Contributed to strategic planning efforts to align AI initiatives with business goals within the AIOps domain.

### TECHNICAL SKILLS:

#### Technical Skills

**Programming Languages:** Python, SQL, Java

**Artificial Intelligence & Generative AI:** OpenAI GPT, Large Language Models (LLMs), Hugging Face Transformers, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, BERT, Semantic Search, Vector Embeddings

**Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, XGBoost, Predictive Analytics, Classification, Regression, Clustering, Time Series Forecasting, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Anomaly Detection

**Natural Language Processing (NLP):** Text Classification, Named Entity Recognition (NER), Sentiment Analysis, Document Summarization, Information Extraction, Conversational AI

**Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Microsoft Excel, KPI Reporting, Dashboard Development, Exploratory Data Analysis (EDA), Statistical Analysis

**Data Engineering:** ETL Pipelines, Apache Kafka, Apache Airflow, Apache Spark, Hadoop, Data Warehousing, Data Modeling, Data Cleaning, Data Validation, Data Governance

**Cloud Platforms:** AWS (S3, EC2, Lambda, SageMaker), Microsoft Azure, Google Cloud Platform (GCP)

**Databases:** MySQL, PostgreSQL, SQL Server, MongoDB, DynamoDB, Azure SQL

**API & Application Development:** FastAPI, Flask, Django, REST APIs, Microservices

**MLOps & DevOps:** MLflow, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Git, Model Deployment, Model Monitoring, Version Control

**Vector Databases:** Pinecone, FAISS

**Tools & Platforms:** Jupyter Notebook, Power Query, DAX, Jira

**Methodologies:** Agile, Scrum, SDLC, Data Privacy, Model Governance, Business Intelligence, Stakeholder Management, Documentation, MIS Reporting

## PROFESSIONAL EXPERIENCE:

### Pfizer

#### Gen AI Engineer:

July 2025-Present

**Description:** Designed, deployed, and optimized scalable AI/ML and generative AI solutions for enterprise applications. Built end-to-end machine learning pipelines, developed predictive models, integrated LLM-based applications into AWS cloud environments, and collaborated with cross-functional teams to automate workflows, improve operational efficiency, and deliver production-ready AI systems.

#### Key Responsibilities:

- Designed and implemented scalable **AI/ML and Generative AI solutions** to solve complex business problems, improve operational efficiency, and support intelligent decision-making across enterprise applications.
- Developed and integrated **Large Language Model (LLM)-based applications using OpenAI GPT and Hugging Face Transformers for document summarization, conversational AI, semantic search, intelligent document processing, and question-answering systems.**
- Implemented **Retrieval-Augmented Generation (RAG)** architectures using **LangChain** and **vector databases such as Pinecone and FAISS** to provide context-aware, accurate, and scalable AI-powered search and knowledge retrieval solutions.
- Processed, cleaned, transformed, and analyzed large-scale enterprise datasets using **Python, SQL, Pandas, and NumPy** to improve data quality and support AI model development.
- Developed **RESTful APIs and FastAPI-based microservices** to integrate AI models into enterprise applications, enabling secure, scalable, and real-time inference.
- Deployed AI and machine learning models on **AWS cloud platforms including SageMaker, EC2, Lambda, and S3** while implementing containerized deployments using Docker and Kubernetes.
- Automated model training, testing, deployment, and monitoring **using MLflow, Jenkins, Git, and CI/CD pipelines**, ensuring reproducible and reliable AI workflows.
- Collaborated closely with **data scientists, software engineers, DevOps teams, product owners, and business stakeholders** to translate business requirements into scalable AI solutions and deliver production-ready applications.
- Led GenAI tool onboarding, developer training, and adoption programs by creating prompt engineering guidelines, best practices, and technical documentation.. Implemented enterprise GenAI developer productivity solutions using **Devin AI, Claude Code, and Cursor AI** to automate code generation, debugging, documentation, and testing activities.
- Designed and deployed an enterprise-grade Agentic AI solution using LangChain and LangGraph with Retrieval-Augmented Generation (RAG), integrating OpenAI GPT-4, Claude, and Llama models with Pinecone/FAISS vector databases for semantic document retrieval; developed scalable FastAPI-based REST services, orchestrated autonomous multi-agent workflows, and deployed the solution on AWS Bedrock, Lambda, and ECS, reducing manual research effort by 70% while improving response accuracy and operational efficiency.

#### Technologies:

**Python, PyTorch, Hugging Face Transformers GPT/LLMs, AWS (S3, Lambda, SageMaker), Pandas, NumPy, SQL, Docker, Kubernetes, REST APIs, Git, Jenkins, CI/CD, Jupyter Notebook.**

### Gen Ai Engineer

Jan 2024-June 2025

#### Walmart:

#### Key Responsibilities

- Designed, developed, and **deployed AI-powered solutions using machine learning, deep learning, and Generative AI** technologies to improve healthcare operations and clinical decision-making.
- Built and optimized **Retrieval-Augmented Generation (RAG) pipelines using LangChain, vector databases, and Large Language Models (LLMs)** to enable intelligent healthcare knowledge retrieval.
- Developed AI applications for document summarization, medical claim analysis, patient support, and

clinical question-answering **using OpenAI GPT models and Hugging Face Transformers.**

- Designed and implemented scalable **data preprocessing pipelines to process structured and unstructured healthcare data using Python, SQL, Pandas, and NumPy.**
- Integrated AI models with **REST APIs and FastAPI-based** microservices for secure and scalable enterprise deployment.
- Implemented **prompt engineering** techniques to improve **LLM response accuracy, reduce hallucinations,** and optimize contextual responses for healthcare use cases.
- Developed semantic search solutions using vector embeddings and **Pinecone/FAISS to enhance enterprise** document retrieval and knowledge management.
- Automated model training, evaluation, deployment, and monitoring **using MLflow, Jenkins, Git, and AWS cloud services, ensuring reproducible AI workflows.**
- Collaborated with **data scientists, software engineers, DevOps teams, and business stakeholders** to deliver scalable AI solutions aligned with organizational objectives.
- **Technologies:**
- **Python, SQL, PyTorch, Hugging Face Transformers, OpenAI GPT, LangChain, RAG, Prompt Engineering, FastAPI, REST APIs, Pandas, NumPy, Pinecone, FAISS, MLflow, AWS (S3, EC2, SageMaker), Docker, Kubernetes, Git, Jenkins, Jupyter Notebook, Vector Databases, Machine Learning, Deep Learning, Generative AI, NLP, MLOps.**

## Jr Data Scientist

Nov 2021-Dec 2022

### Wells Fargo

#### Responsibilities:

- Performed **exploratory data analysis (EDA)** and statistical analysis on large structured and unstructured datasets using **Python, Pandas, NumPy, and SQL** to identify patterns, trends, and actionable business insights.
- Implemented machine learning models using **scikit-learn, TensorFlow, and PyTorch for tasks like predictive analytics, classification, and regression,** resulting in a 20% improvement in model accuracy.
- Built end-to-end **data science workflows** including data collection, cleaning, preprocessing, feature engineering, model training, validation, and performance optimization.
- Collaborated with data scientists to optimize **data pipelines and pre-process data for model training,** reducing model training time by 25%.
- Performed **hyperparameter tuning** using Grid Search, Random Search, and cross-validation techniques to improve model performance and generalization.
- Developed predictive models for business use cases including **customer behavior analysis, risk prediction, fraud detection, forecasting, and anomaly detection.**
- Worked with **relational (MySQL, PostgreSQL) and NoSQL (MongoDB, DynamoDB) databases** to efficiently store and retrieve structured and unstructured data for model training and analysis.
- Developed interactive dashboards **using Power BI and Tableau, providing stakeholders** with actionable insights from complex datasets, improving data accessibility and visualization by 40%.
- Collaborated with **data engineers, business analysts, software engineers, and stakeholders** to define analytical requirements and deliver data-driven solutions.
- Implemented feature engineering techniques to improve the predictive **power of machine learning models, boosting model** accuracy by 15%.

#### Technologies:

Python, SQL, TensorFlow, PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, FastAPI, REST APIs, AWS (S3, EC2, Lambda, SageMaker), Microsoft Azure, Google Cloud Platform (GCP), MySQL, PostgreSQL, MongoDB, Apache Spark, Power BI, Tableau, Matplotlib, Seaborn, MLflow, Git, Jupyter Notebook, Feature Engineering, Predictive Analytics, Classification, Regression, Machine Learning, Deep Learning, Agile, MLOps.

## Data Analyst

May 2019-Oct 2021

### ZelisHealthCare

#### Responsibilities:

- Gathered, analyzed, and interpreted data from multiple internal and external sources to support business decision-making and operational improvements.
- Designed, developed, and maintained ETL processes to **extract, clean, transform, and load data into centralized data warehouses**, ensuring high data quality and consistency.
- Wrote and optimized complex **SQL queries, stored procedures, views, and joins** to extract actionable insights, generate reports, and support analytical requirements.
- Utilized **Python (Pandas, NumPy, Matplotlib, and Seaborn)** for data preprocessing, exploratory data analysis (EDA), statistical analysis, automation, and reporting.
- Built interactive dashboards, scorecards, and **executive reports using Power BI to visualize KPIs, business performance, trends, and operational metrics**.
- Performed **data validation, quality assurance testing, and root cause analysis** to identify data inconsistencies and improve overall data accuracy.
- Conducted weekly data quality review meetings with cross-functional teams to discuss **data issues, implement corrective actions, and enhance reporting accuracy**.
- **Designed and published operational, financial, and performance reports** to provide timely business insights for stakeholders and leadership teams.
- **Developed Python and SQL automation** scripts to streamline repetitive reporting tasks, reducing manual effort and improving operational efficiency.
- Managed **Management Information System (MIS)** reporting activities by delivering accurate, timely, and standardized reports for business operations.
- Created self-service **dashboards and reporting solutions that enabled business users to access real-time** insights and make informed decisions.
- **Defined, monitored, and tracked data quality metrics, KPIs, and validation rules** to ensure consistent, reliable, and high-quality analytical outputs.
- Collaborated **with business stakeholders, product owners, data engineers, and technical teams** to gather reporting requirements and deliver customized analytical solutions.
- Prepared comprehensive documentation for **ETL workflows, SQL queries, data models, dashboards, and reporting processes to ensure maintainability, reproducibility, and knowledge sharing**.

#### Technologies:

Python, SQL, Power BI, Tableau, Pandas, NumPy, Matplotlib, Seaborn, Microsoft Excel (Advanced), ETL, Data Warehousing, MySQL, PostgreSQL, SQL Server, Azure SQL, Power Query, DAX, Data Modeling, Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, KPI Reporting, Dashboard Development, Data Visualization, Automation, Jupyter Notebook, MIS Reporting, Agile, Jira.

#### Education:

Master of Science in Computer Science

Northern Arizona University | May 2024

**Bachelor of Science in Information Technology**

Anurag Group Of Institutions, India | May 2019